Rundong Zhou

Curriculum Vitae

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RESEARCH INTERESTS

• Turbulent convection • Geophysical flows • Stability analysis • Phase transition • Numerical methods

EDUCATION

Ph.D. October 2024 - Present
University of Twente, **Physics of Fluids department** Enschede, Netherlands
Thesis: Instabilities, transitions, and turbulence in geophysical flows
Supervisors: Prof. Detlef Lohse and Prof. Roberto Verzicco

M.Sc. September 2024
Chalmers University of Technology Gothenburg, Sweden
Joint with the **University of Gothenburg**, **Department of Physics**
Major in Complex Adaptive Systems

Erasmus+ Exchange Program August 2023 - July 2024
University of Twente, **Physics of Fluids department** Enschede, Netherlands
Thesis: Swirling Kolmogorov flow
Supervisors: Dr. Chris Howland and Prof. Detlef Lohse

B.A.Sc. in Engineering Science June 2021
University of Toronto, **Division of Engineering Science** Toronto, Canada
Major in Engineering Physics
Thesis: Spectral methods in polar coordinates with discrete Hankel transforms
Supervisor: Prof. Nicolas Grisouard

EXPERIENCES

Visiting Ph.D. student September 2026
Max Planck Institute for Dynamics and Self-Organization Göttingen, Germany
Theory of Turbulent Convection Group
Supervisor: Prof. Olga Shishkina

LIST OF PUBLICATIONS

Zhou, R., Verzicco, R., Shishkina, O., and Lohse, D. (2026). Onset of convection in a centrifuge tube with a radial temperature gradient. Under reiview for *Journal of Fluid Mechanics*.

Zhou, R., Lefauve, A., Verzicco, R. and Lohse, D. (2026). Transition to the ultimate regime of turbulent convection in stratified inclined duct flow. *Physical Review Fluids, 11*(4), 044802. doi.org/10.1103/psz6-f48t

Zhou, R. and Grisouard, N. (2023). Spectral solver for Cauchy problems in polar coordinates using discrete Hankel transforms. Preprint. arXiv:2210.09736

TEACHING

Advanced Fluid Mechanics course September - November 2026
University of Twente Enschede, Netherlands
Lectures, tutorial, assistance, and grading of assignments.

Turbulence course November 2025 - Janurary 2026
University of Twente Enschede, Netherlands
Tutorial, assistance, and grading of assignments.

PEER REVIEWS

Physical Review Letters (1), *Journal of Fluid Mechanics* (3), *Physical Review Fluids* (1), *Physical Review Research* (1), *International Journal of Heat and Fluid Flow* (1).

CONFERENCES & TRAINING

14th International Symposium on Turbulence and Shear Flow Phenomena July 2026
Heidelberg, Germany

Conference proceeding and oral presentation on *Boundary layers and transport in stratified inclined duct flow.*

10th International Conference on Rayleigh-Bénard Turbulence June 2026
École Normale Supérieure de Lyon Lyon, France
Poster presentation of Melting an ice cylinder in a rotating frame: Thermal boundary layer instabilities induced morphology.

Fall school in turbulence, Bridges over turbulent matters October 2025
Institut d'Études scientifiques de Cargèse, Université Côte d'Azur Corsica, France
Bridges between the various facets of contemporary research on turbulence, from mathematical fluids to experiments, from phenomenology to stochastic physics. Lectures on renormalization group, Lagrangian turbulence, quantum & wave turbulence, geophysical turbulence, etc.

2nd European Fluid Dynamics Conference August 2025
University College Dublin Dublin, Ireland
In person attendance with talk given on *Transition to the ultimate regime of turbulent convection in stratified inclined duct flow.*

CMiF summer school 2025 June 2025
CAES du CNRS, La Vieille Perrotine Oléron, France
Complex motion in fluids, from micro to macroscopic. Lectures on nano fluids, sports physics, singularities, multiphase flows, planetary flows, etc.

JMBC – Multiphase flow and phase transitions course April 2025
J.M. Burgerscentrum for Fluid Mechanics, University of Twente Enschede, Netherlands
Lectures on melting, solidification and crystallization, immersed boundary method, phase field method, bubble dynamics, etc.

Max Planck Center Meeting April 2025
Max Planck Institute for Dynamics and Self-Organization Göttingen, Germany
Joint meeting program of Max Planck Center for Complex Fluid Dynamics. In person attendance with talk given on *Transition to the ultimate regime of turbulent convection in stratified inclined duct flow.*

CISM-EUROMECH Advanced School April 2025
International Centre for Mechanical Sciences Udine, Italy
Instabilities in turbulence. Lectures on routes to turbulence, turbulent bifurcations, nonlinear instabilities, and Rayleigh-Bénard turbulence.

Summer school in mathematics June 2023
Université Grenoble Alpes, Institut Fourier Grenoble, France
Topics in new trends in mathematical fluid mechanics. Lectures on mathematical analysis of incompressible fluid flow, geophysical flows, theory of water waves, singular solutions of the Euler or Navier-Stokes equations, etc.

FEATURED COURSES

Toronto: Continuum Mechanics, Computational Physics, Nonlinear Physics, Statistical Mechanics, Groups and Symmetries
Chalmers: Dynamical Systems, Non-equilibrium Processes in Physics Chemistry and Biology, Quantum Field Theory, Artificial Neural Networks
Twente: Advanced Fluid Mechanics, Turbulence, Functional Analysis, Mathematical and Numerical Physics

PROGRAMMING SKILLS

Languages: Python, Fortran, LaTeX, Matlab, Mathematica, C
Libraries: Dedalus, AFiD, GHOST, NumPy, Matplotlib, SciPy

Latest Update: October 15, 2026